

## ARM7TDMI – TSMC 0.18μm

### Features:

- Full Custom 32/16-bit RISC architecture (ARM v4T)
- 32-bit ARM instruction set for maximum performance
- 16-bit Thumb instruction set for increased code density
- Unified bus interface, 32-bit data bus carries both instructions and data
- Three-stage pipeline
- 32-bit ALU and high-performance multiplier
- Fully static operation
- Coprocessor interface
- Extensive debug facilities:
  - EmbeddedICE-RT real-time debug unit
  - JTAG interface unit
  - Interface for direct connection to Embedded Trace Macrocell

### Summary

#### ■ Silicon vendor

- TSMC

#### ■ Process name/geometry

- TSMC 0.18μm “Generic” CL018 1.8V/3.3V

#### ■ Layers Supported

- 1 PolySilicon, 4, 5 or 6 Metal process

#### ■ Area of macrocell

- 0.53mm<sup>2</sup>

#### ■ Process Conditions

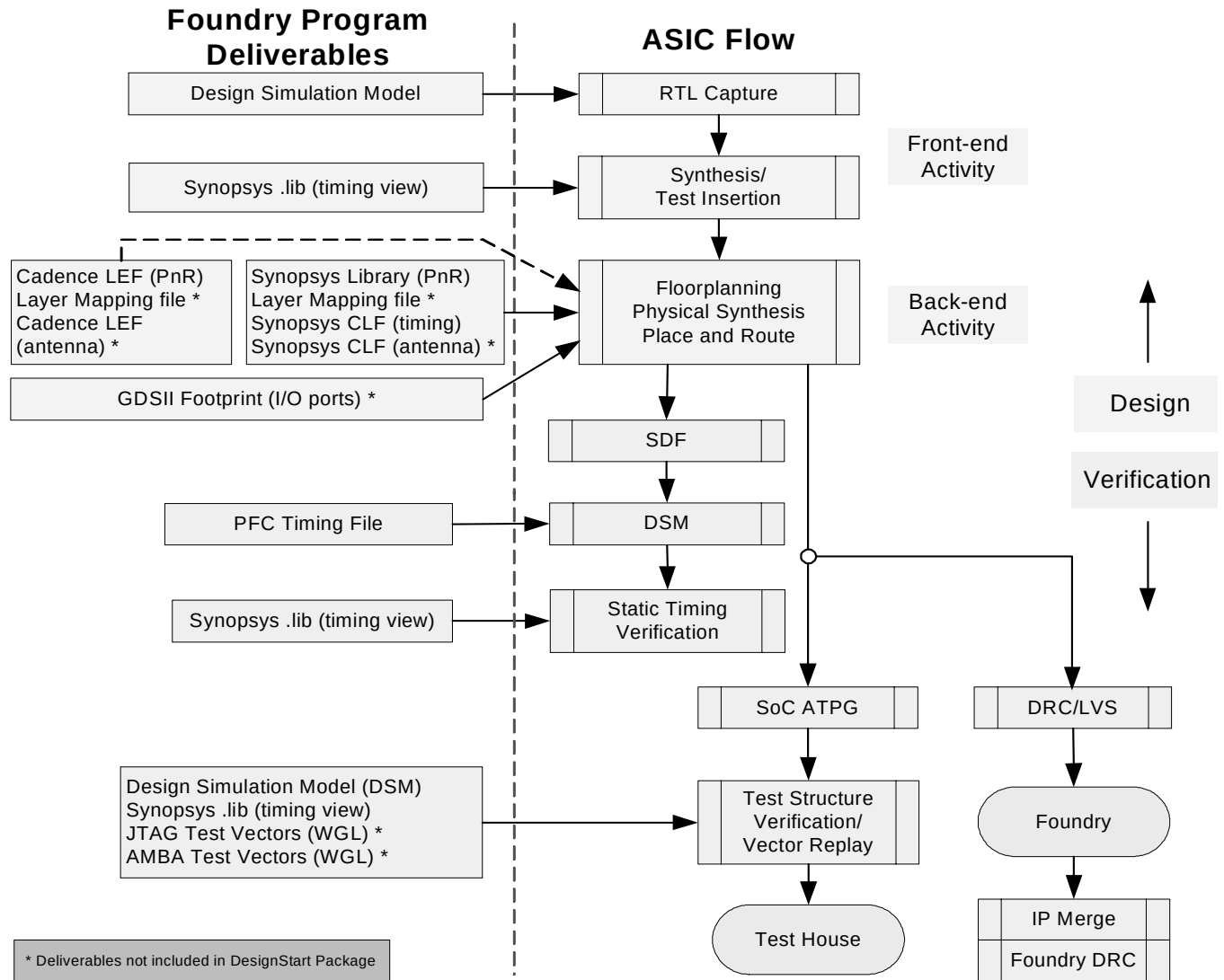
- worst case
  - +125°C, 1.62V
- best case1
  - 0°C, 1.98V
- best case2
  - -40°C, 1.98V

### Configuration, Performance and Documentation

| Parameter                                | Value                                       |
|------------------------------------------|---------------------------------------------|
| Module name                              | ARM7TDMI                                    |
| Module revision                          | Rev3                                        |
| I-Cache Size                             | N/A                                         |
| D-Cache Size                             | N/A                                         |
| Worst Case Max. Clock Frequency          | 89 MHz                                      |
| Power Figure (DhryStone 2.1)             | 0.241 mW/Mhz                                |
| Size of macrocell                        | 0.92 (w) x 0.58 (h) mm                      |
| Number of metal layers used in core      | 4 metal layers                              |
| Metal layers free for routing above core | Metals 5 and 6                              |
| Restriction on Rotation                  | None - The core can be rotated as required. |

| Parameter                                           | Value                                                            |
|-----------------------------------------------------|------------------------------------------------------------------|
| Test Methodology                                    | - AMBA Parallel (SIM) vectors<br>- JTAG Serial Vectors           |
| Number of Chip I/O pins needed for ARM core testing | 5 (minimum for JTAG)                                             |
| Number of Signal pins                               | 212 (distributed around 3 sides of the core Top: M2, Sides : M3) |
| Number of Vdd/Vss pairs                             | 8 (all must be routed)                                           |
| AMBA Interface                                      | AHB wrapper available                                            |

| Document                   | Document Number   |
|----------------------------|-------------------|
| Technical Reference Manual | ARM DDI 0029      |
| Integration Manual         | ARM DDI 0101      |
| AHB Wrappers TRM           | ARM DDI 0169      |
| DesignStart Kit User Guide | PR224-PRDC-006299 |
| Performance Summary        | GENC-006013       |



## Inventory

### Documentation

- Release Note
- Technical Reference Manual
- Performance Summary
- Macrocell Integration Manual
- DesignStart Kit User Guide
- AHB Wrappers User Guide

### Simulation Environment

- AMBA Simulation Environment
- JTAG Simulation Environment

### RTL

- AMBA AHB Wrapper verilog

### Synthesis

- AHB Wrapper scripts

### Design Simulations Models

- Solaris & Linux compiled
  - Cadence: NC-Sim
  - Synopsys: VCS
  - Mentor: ModelSim

### Timing views

- Synopsys .lib
- Synopsys .CLF
- I/O slow & fast corners

### Layout (data)

- Synopsys Library .FRAM
- Cadence Library .LEF

### Vectors

- Example AMBA .sim test vectors
- Example JTAG .vec test vectors